

CogAge all-subject optimizer comparison

Classes and availability

Bending=0, Walking=1, Sitting=2, Standing=3. Four-class optimizer runs include S1, S2, S5, and S7. S3, S6, and S8 are previewed but excluded because class pairs are missing; S4 has no original test folders.

Data and splits

Each window is 128 x 33. Original _1_extracted folders supply train; _2_extracted folders supply held-out test. Validation Events are selected only from each subject's original train windows, with Event groups kept intact. A new StandardScaler is fitted only on model-training windows. No class balancing is used.

Network and fixed settings

NumPy MLP: 4224 -> 32 -> 16 -> 4 logits, ReLU hidden layers, 135,796 parameters, stable sparse cross-entropy. Seed 42; 100 epochs; LR 0.01; mini-batch 8; momentum 0.9; RMSprop decay 0.99; Adam betas (0.9,0.999); AdamW weight decay 0.01.

Optimizer updates

Full-batch GD updates once per epoch; mini-batch once per batch; stochastic GD once per example. Momentum accumulates velocity; Adagrad accumulates squared gradients; RMSprop uses an exponential squared-gradient average; Adam uses first and second moments; AdamW decouples weight decay. Compare update counts and elapsed time, not epochs alone.

Limits and disclosure

Few Events make validation estimates noisy. These are within-subject results, not participant-independent generalization. The raw-file audit, interpolation/removal counts, and discarded windows are in each subject dataset folder. GitHub Copilot assisted with drafting/debugging; verify and disclose as required.

Subject	Optimizer	Val loss	Epoch	Test acc.	Macro F1	Seconds	Updates
S1	Full-batch GD	1.001	100	0.850	0.842	0.95	100
S1	Mini-batch GD	1.125	2	0.800	0.783	1.28	400
S1	Stochastic GD	0.002	1	0.825	0.827	7.10	3,200
S1	SGD + Momentum	0.000	10	0.825	0.825	2.09	400
S1	Adagrad	0.043	1	0.950	0.948	2.06	400
S1	RMSprop	3.454	5	0.900	0.896	2.75	400
S1	Adam	2.242	1	0.900	0.893	3.39	400
S1	AdamW	2.240	1	0.900	0.893	3.30	400
S2	Full-batch GD	2.106	3	0.750	0.683	0.53	100
S2	Mini-batch GD	1.859	4	0.800	0.757	1.26	400
S2	Stochastic GD	0.502	3	0.975	0.975	8.32	3,200
S2	SGD + Momentum	1.282	3	0.825	0.825	2.09	400
S2	Adagrad	0.043	1	0.950	0.948	2.06	400
S2	RMSprop	3.454	5	0.900	0.896	2.75	400
S2	Adam	2.242	1	0.900	0.893	3.39	400
S2	AdamW	2.240	1	0.900	0.893	3.30	400
S5	Full-batch GD	0.618	85	0.800	0.750	0.60	100
S5	Mini-batch GD	0.176	74	0.775	0.705	0.98	400
S5	Stochastic GD	0.570	6	0.875	0.875	5.93	3,200
S5	SGD + Momentum	0.000	10	0.750	0.661	1.55	400
S5	Adagrad	0.054	100	0.850	0.849	1.98	400
S5	RMSprop	0.000	2	0.625	0.577	2.63	400
S5	Adam	0.006	4	0.750	0.723	3.51	400
S5	AdamW	0.006	4	0.750	0.723	3.74	400
S7	Full-batch GD	0.138	100	0.615	0.540	0.51	100
S7	Mini-batch GD	0.057	100	0.667	0.588	1.04	400
S7	Stochastic GD	0.000	100	0.744	0.661	6.31	3,000
S7	SGD + Momentum	0.063	1	0.641	0.572	1.44	400
S7	Adagrad	0.000	100	0.718	0.642	2.18	400
S7	RMSprop	-0.000	1	0.744	0.696	2.86	400
S7	Adam	0.000	3	0.744	0.661	3.10	400
S7	AdamW	0.000	3	0.744	0.661	3.10	400

Validation loss curves

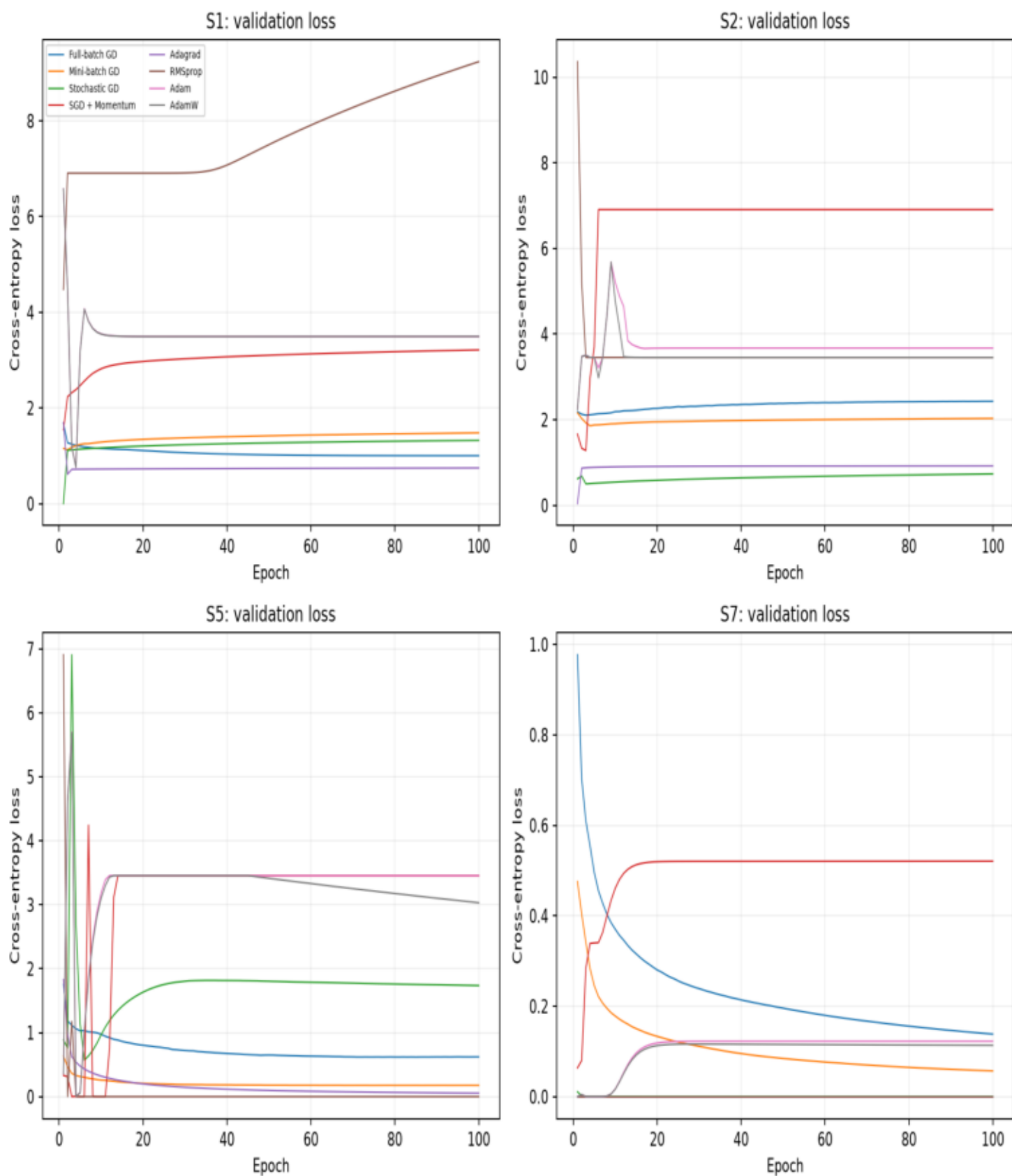


Figure 1. Validation loss by epoch for all eight optimizers, shown separately for S1, S2, S5 and S7.

Validation accuracy curves

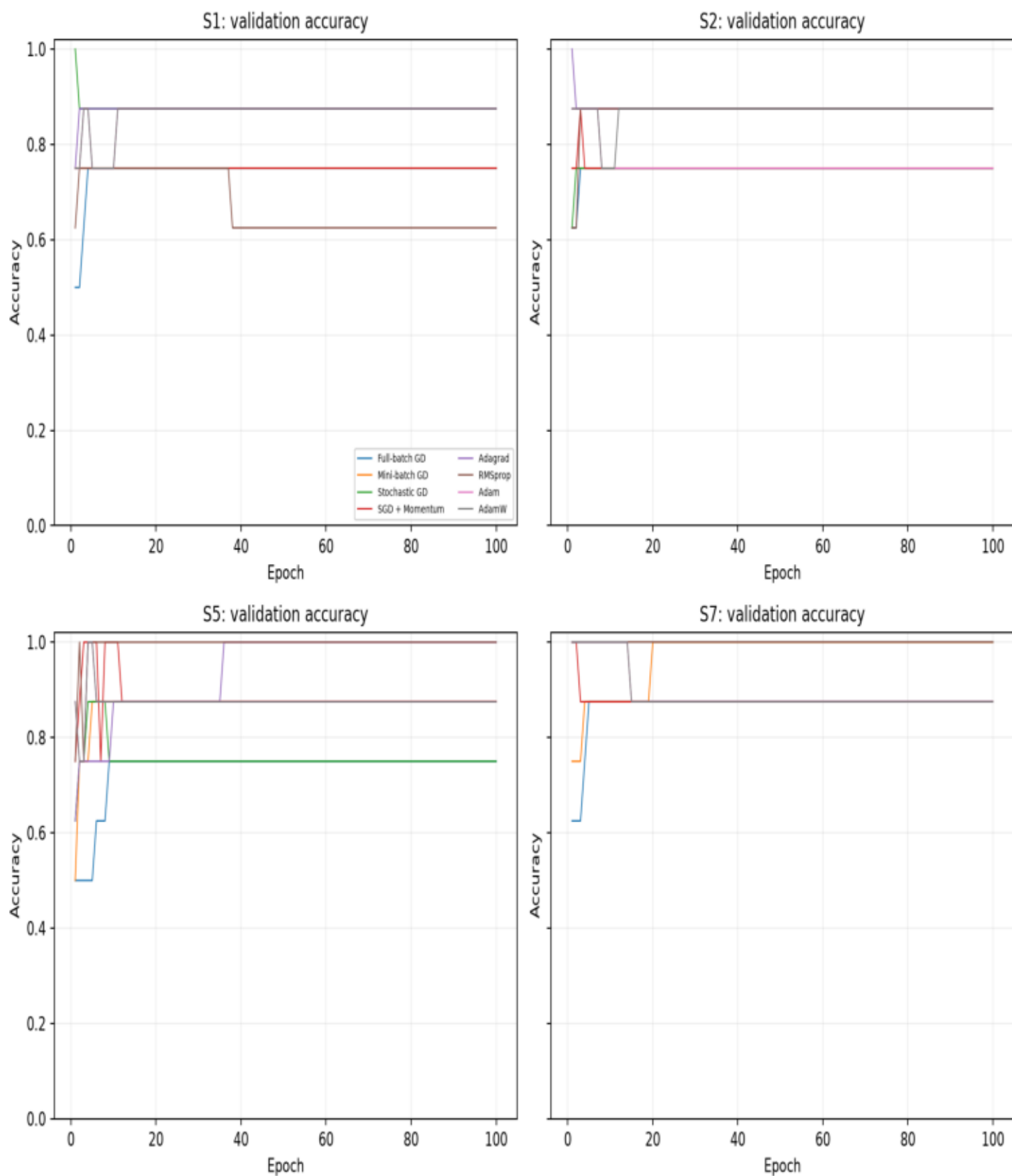


Figure 2. Validation accuracy by epoch. Small validation sets make each misclassification consequential.

Convergence and test metrics

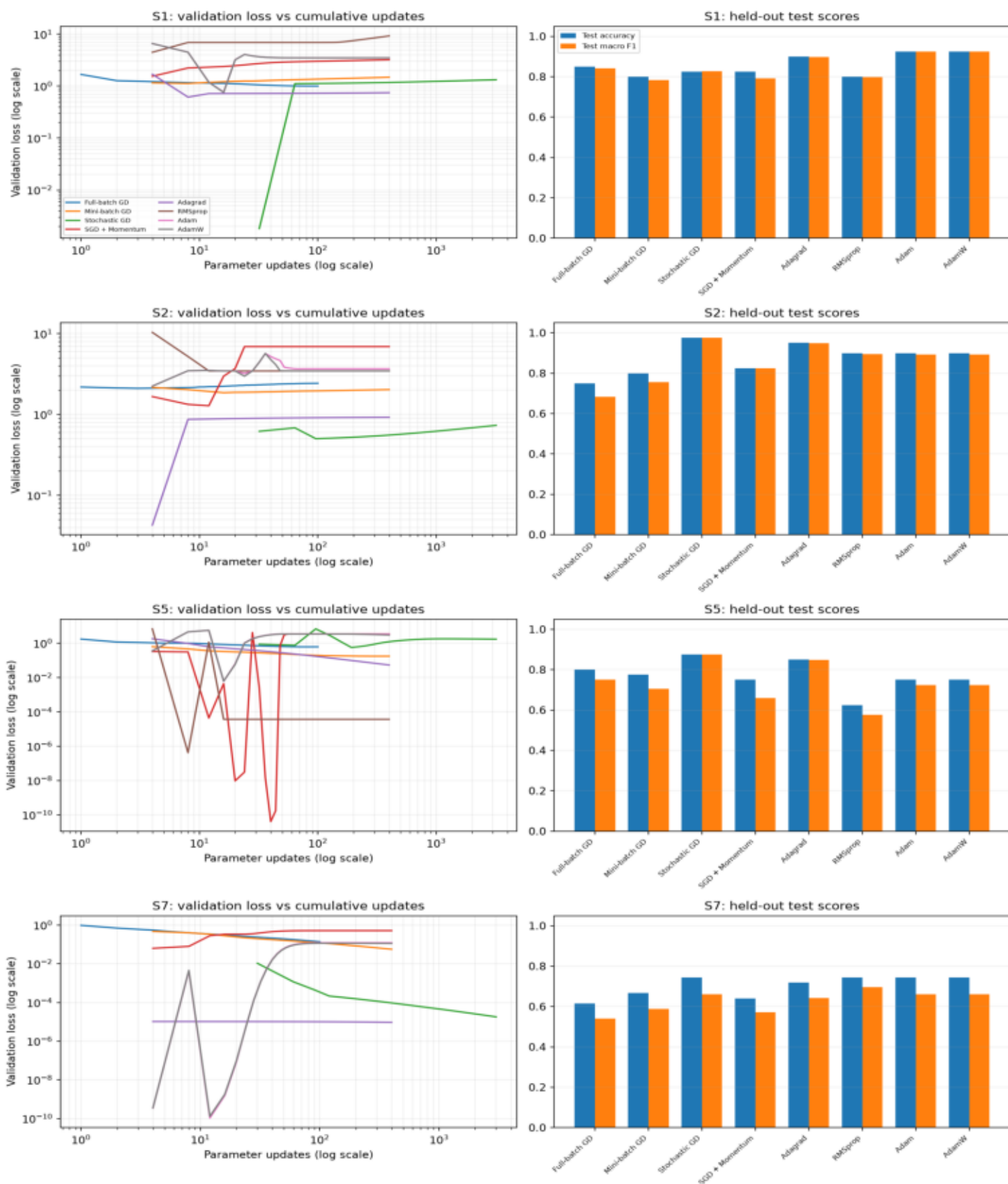
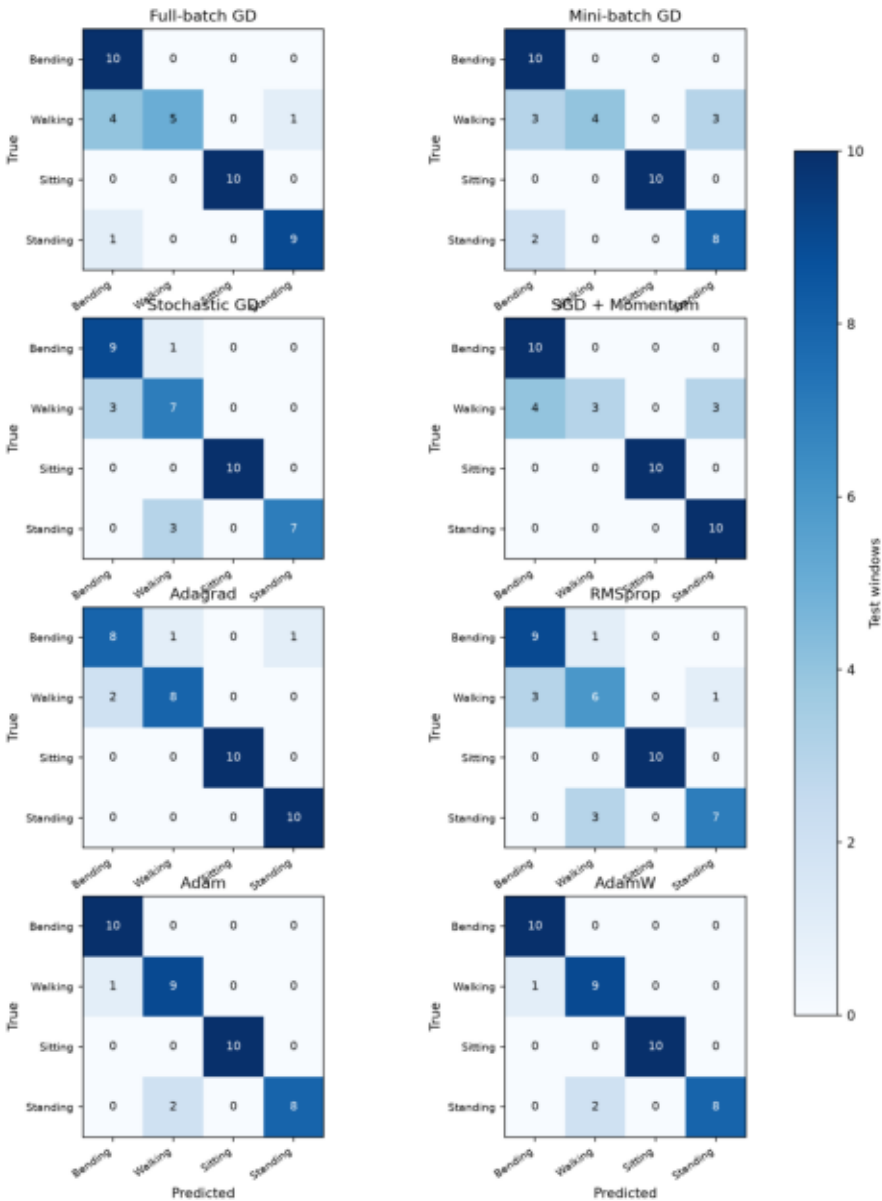


Figure 3. Validation loss versus cumulative updates and held-out test accuracy/macro F1 by subject. Update counts differ by batch size.

Held-out test confusion matrices: S1, S2

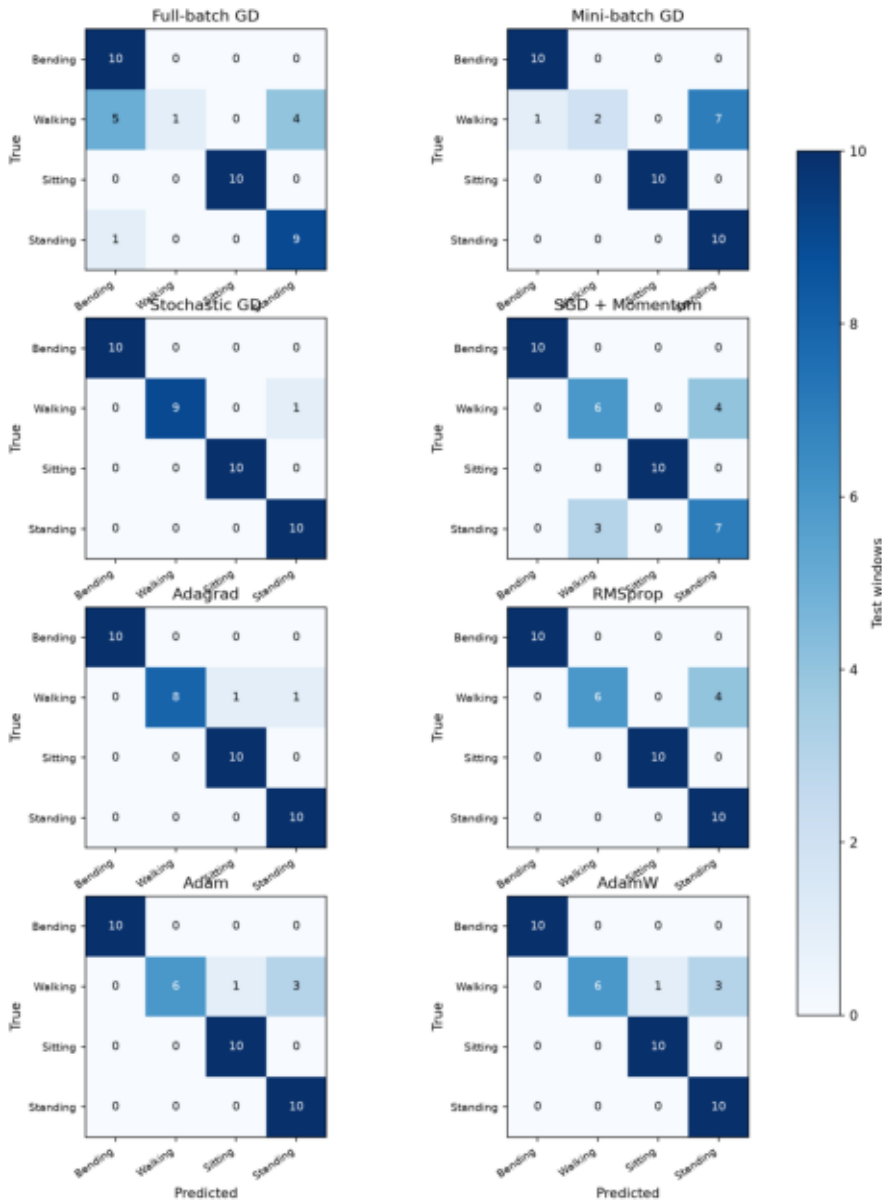
S1

S1: held-out test confusion matrices (rows=true, columns=predicted)



S2

S2: held-out test confusion matrices (rows=true, columns=predicted)

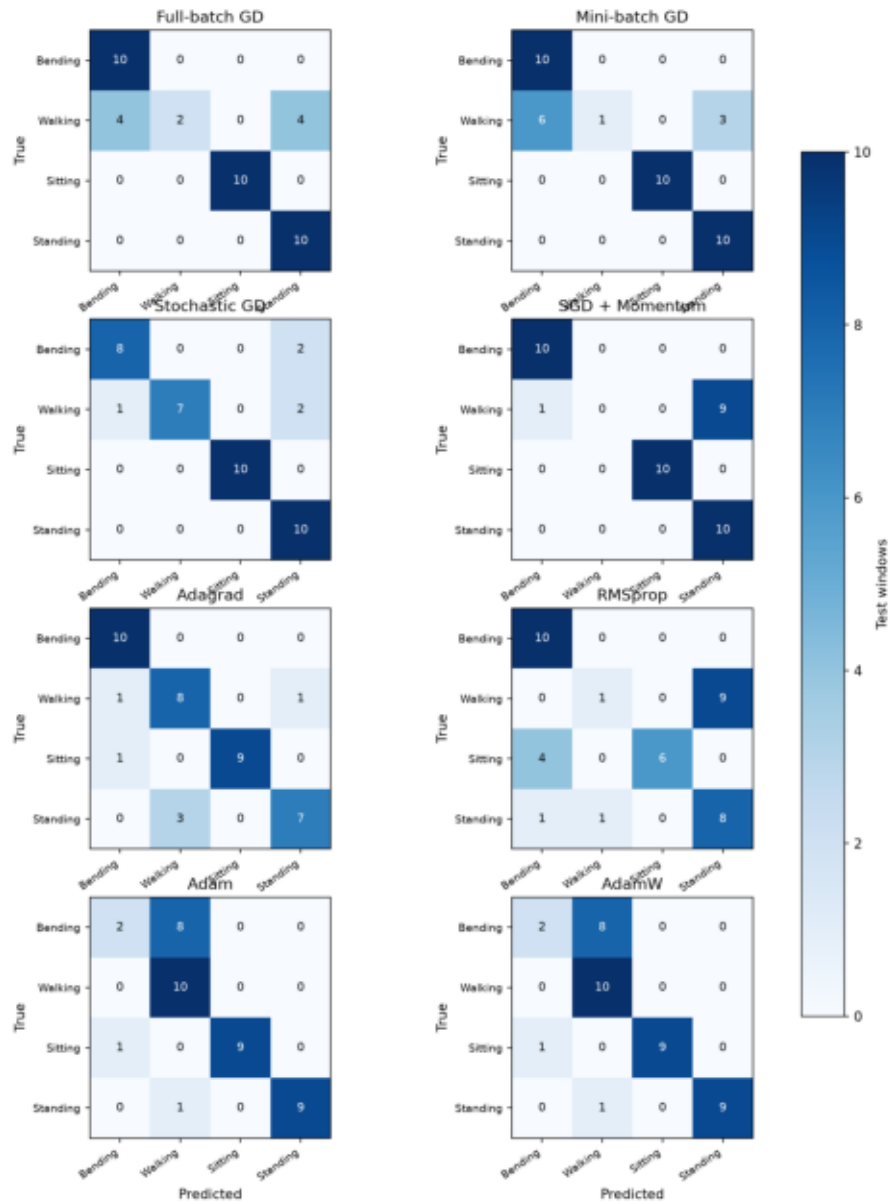


Each matrix is 4 x 4; rows=true, columns=predicted; labels: Bending, Walking, Sitting, Standing.

Held-out test confusion matrices: S5, S7

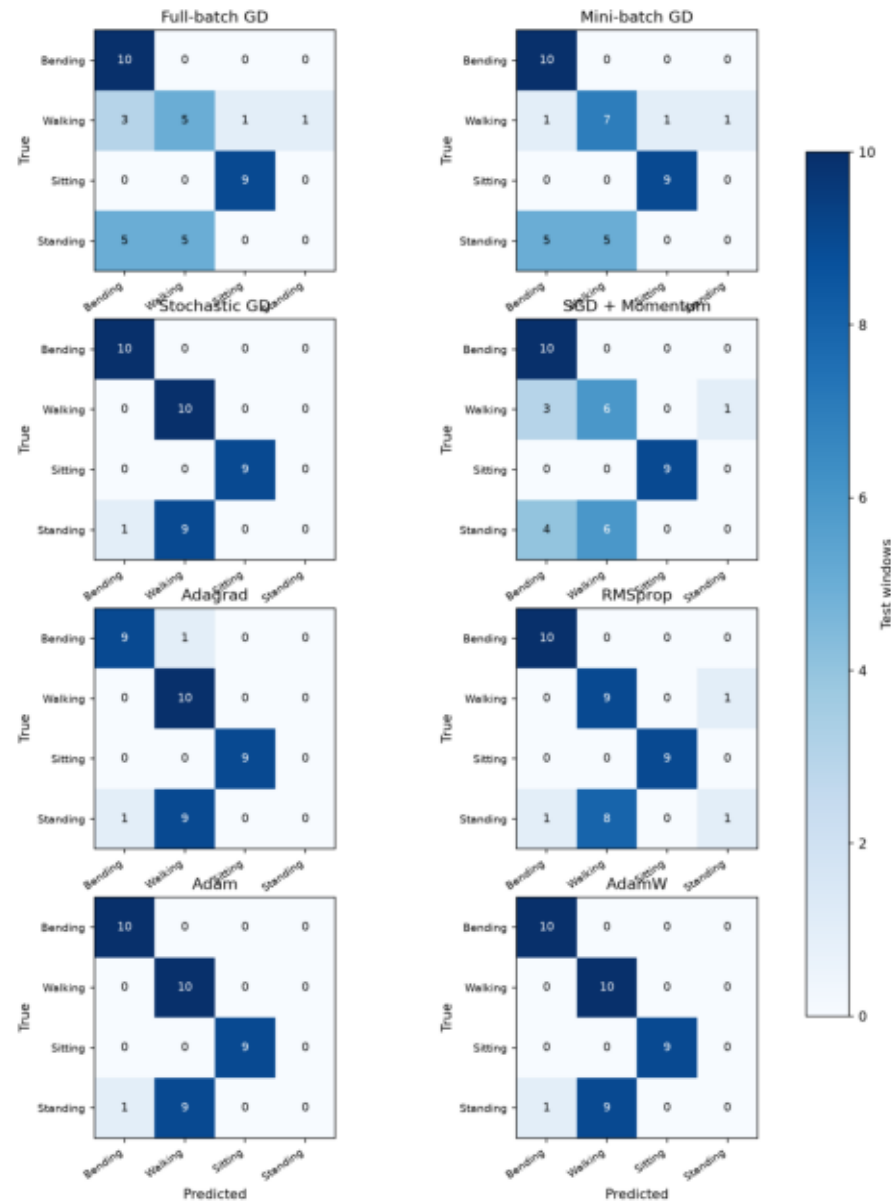
S5

S5: held-out test confusion matrices (rows=true, columns=predicted)



S7

S7: held-out test confusion matrices (rows=true, columns=predicted)



Each matrix is 4 x 4; rows=true, columns=predicted; labels: Bending, Walking, Sitting, Standing.